



Procyon: Promoting Fine-Grain Multi-Tenancy to Optimize Sparse Streaming Accelerators

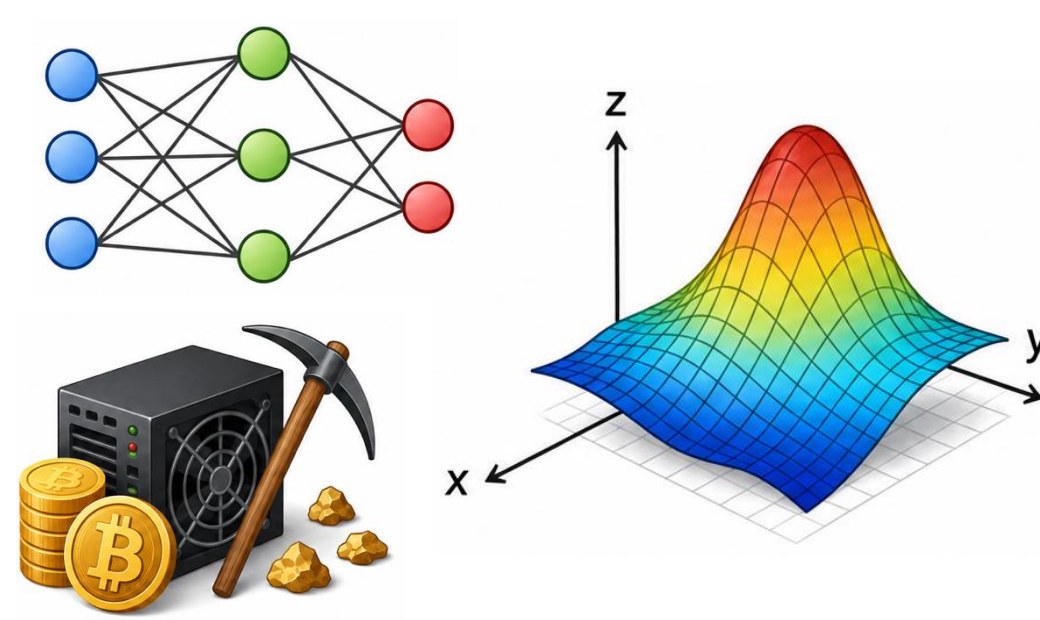
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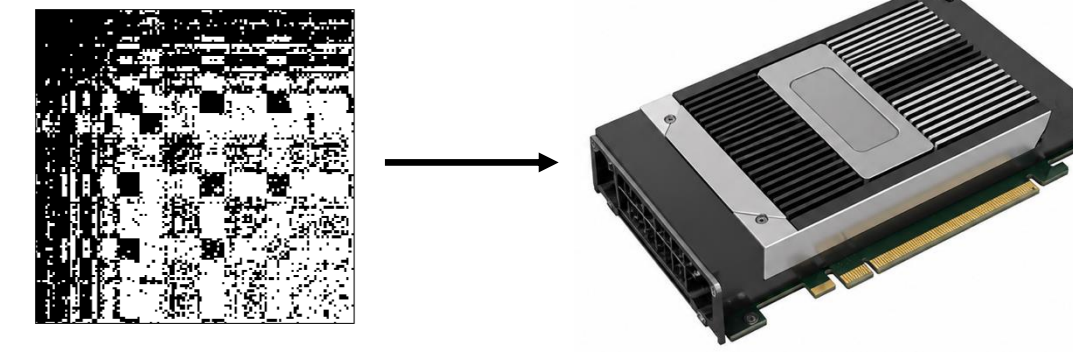
Introduction



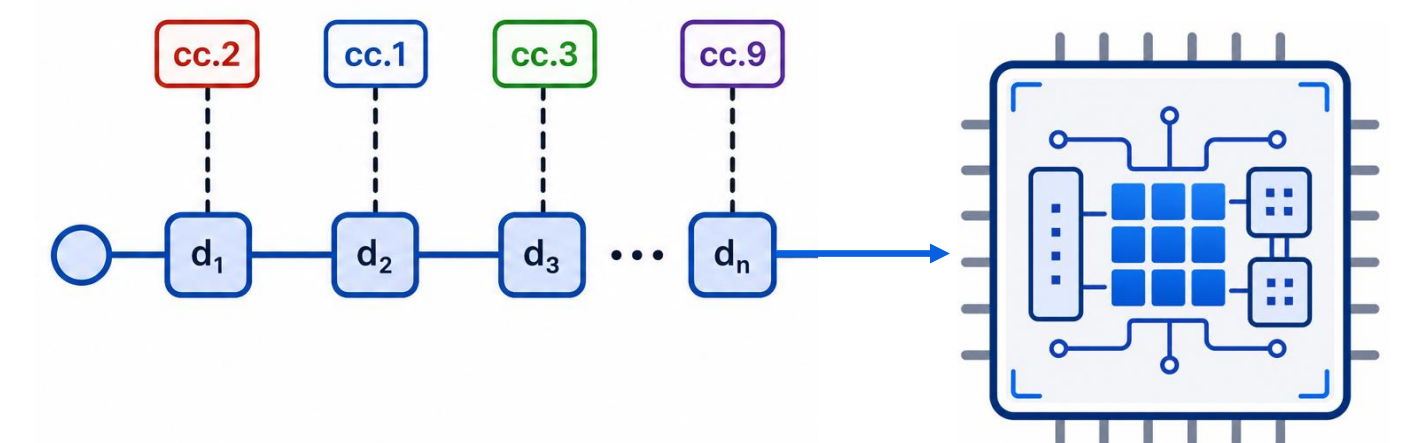
Moore's Law drove decades of computing scalability



Modern sparse workloads expose the limits of traditional processors



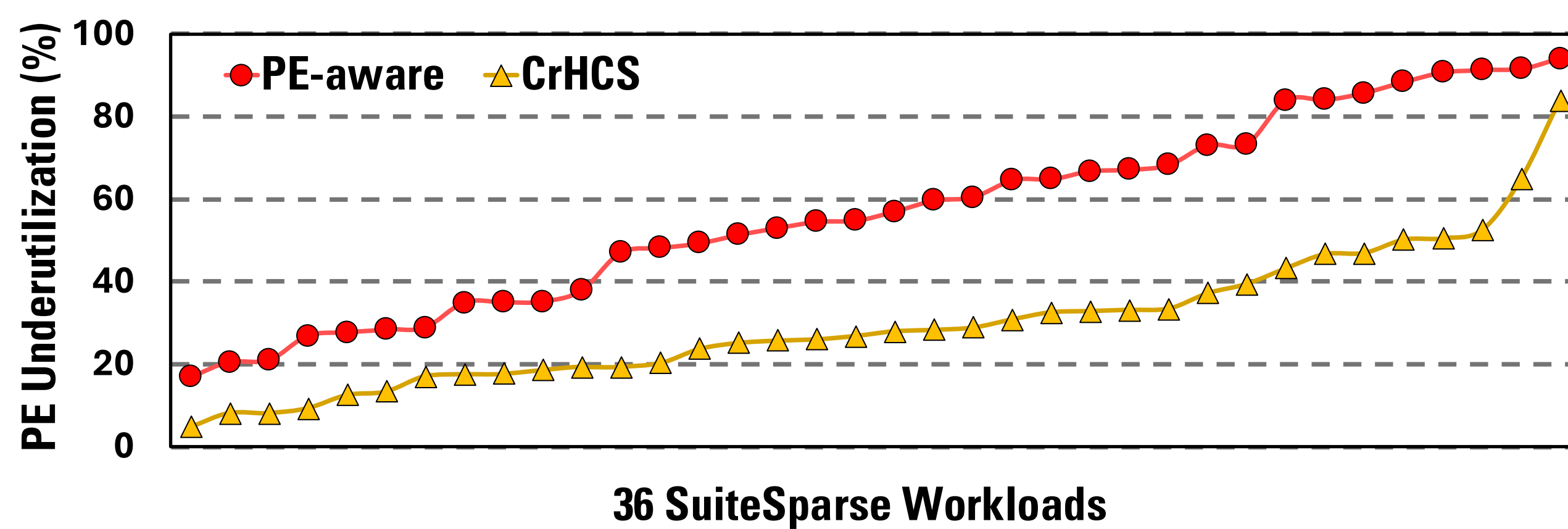
Streaming accelerators are proving highly effective for sparse workloads.



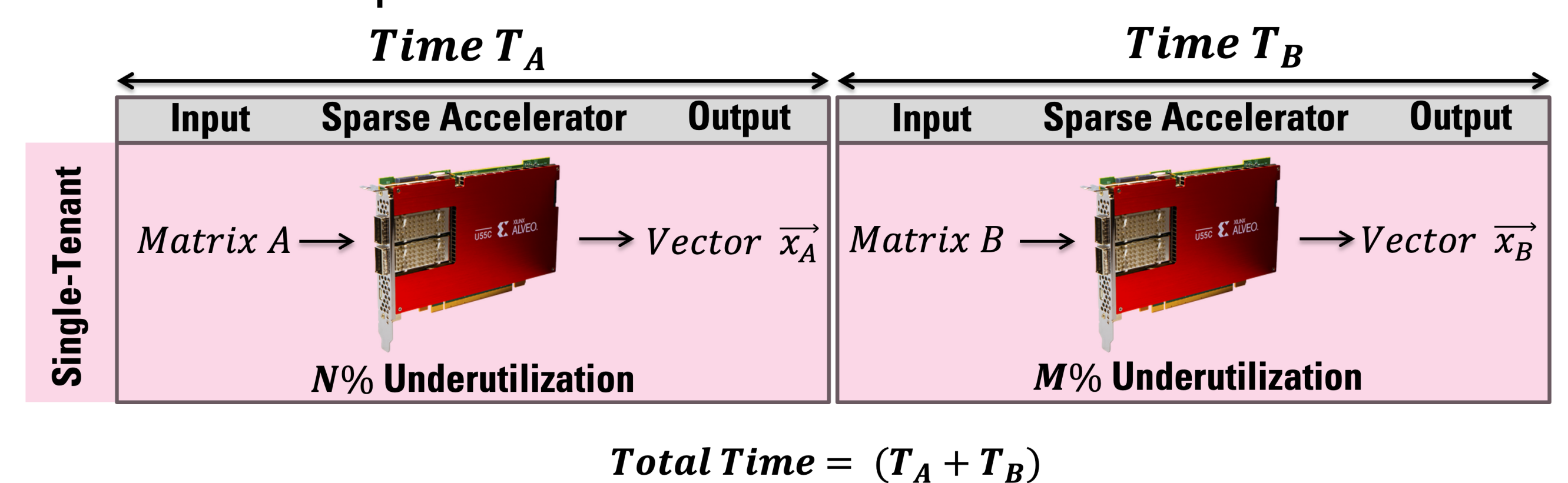
Data scheduling directly affects their performance and execution model

Challenges

1. Poor data scheduling leaves the PEs underutilized

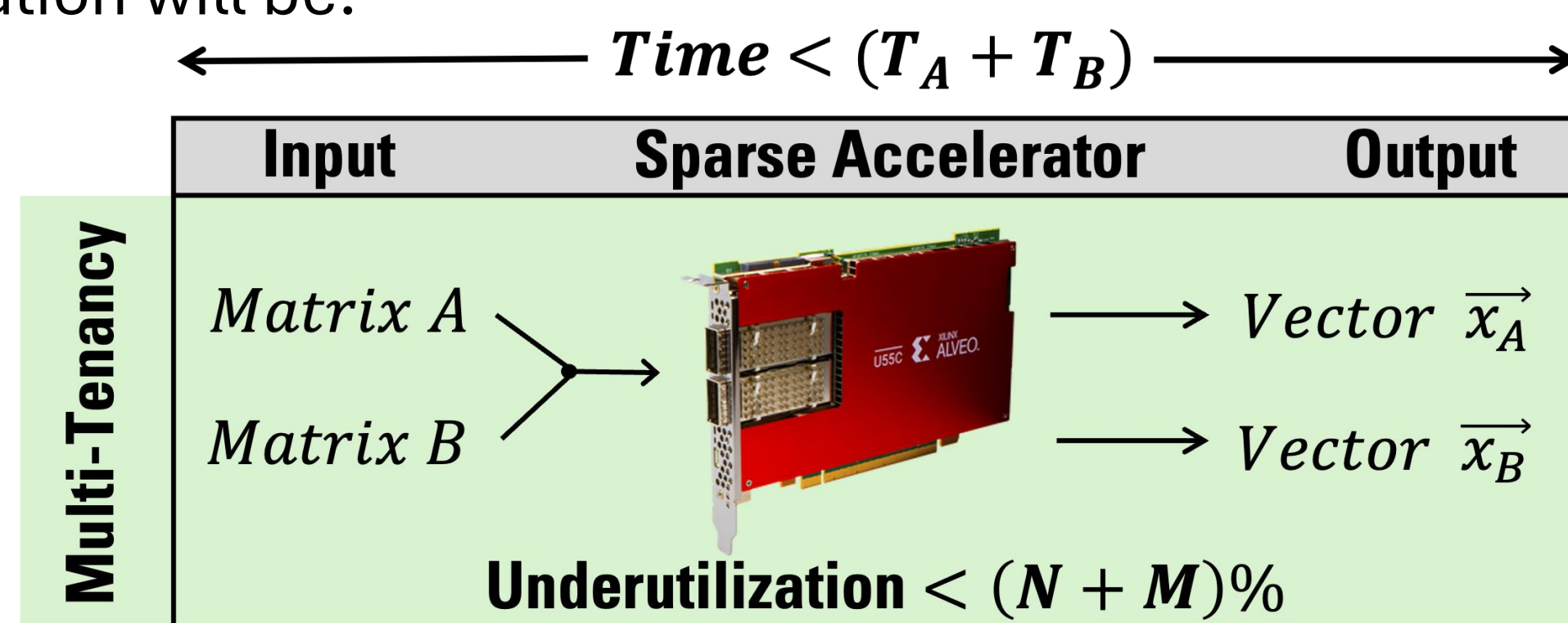


2. Single workload execution model fails to concurrently execute multiple workloads



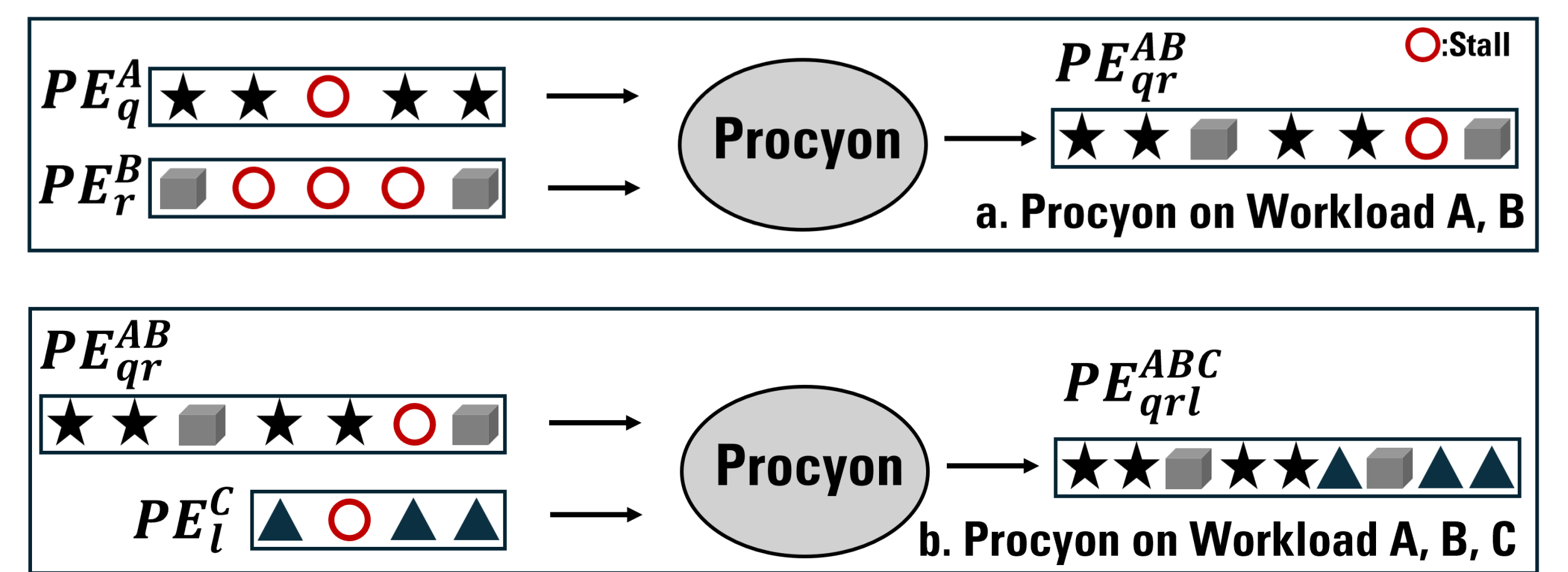
Key Insight

If we fuse the PE instructions of multiple workloads into a unified set of instructions, it enables multi-tenant execution of a PE while eliminating PE stalls by filling them with instructions from other workloads. Essentially, the execution model of a streaming accelerator doing sparse matrix-vector multiplication will be:



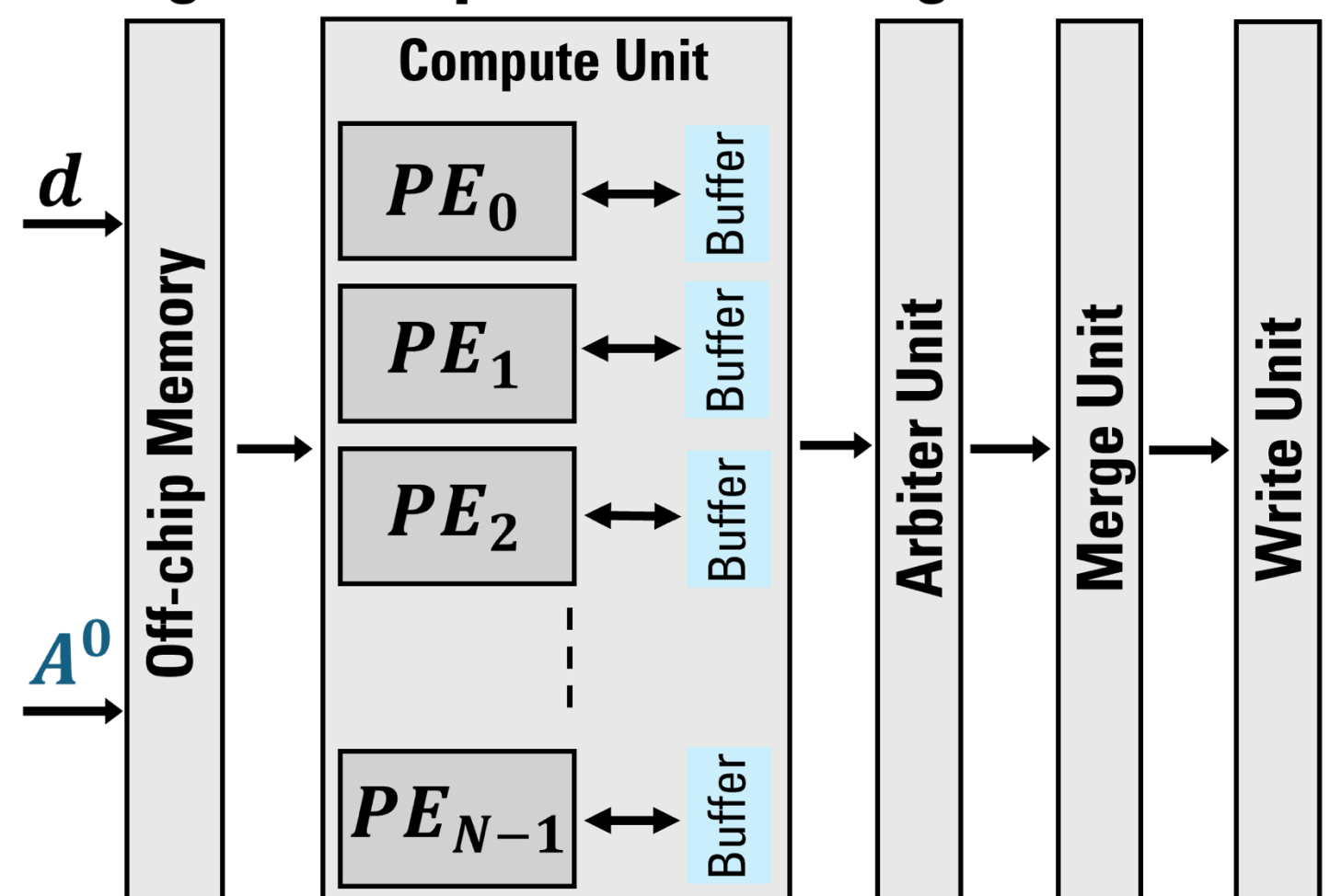
Procyon Scheduler

q^{th} , r^{th} and l^{th} PE datalist of workload A, B and C respectively are fused together into one datalist using Procyon in two steps:

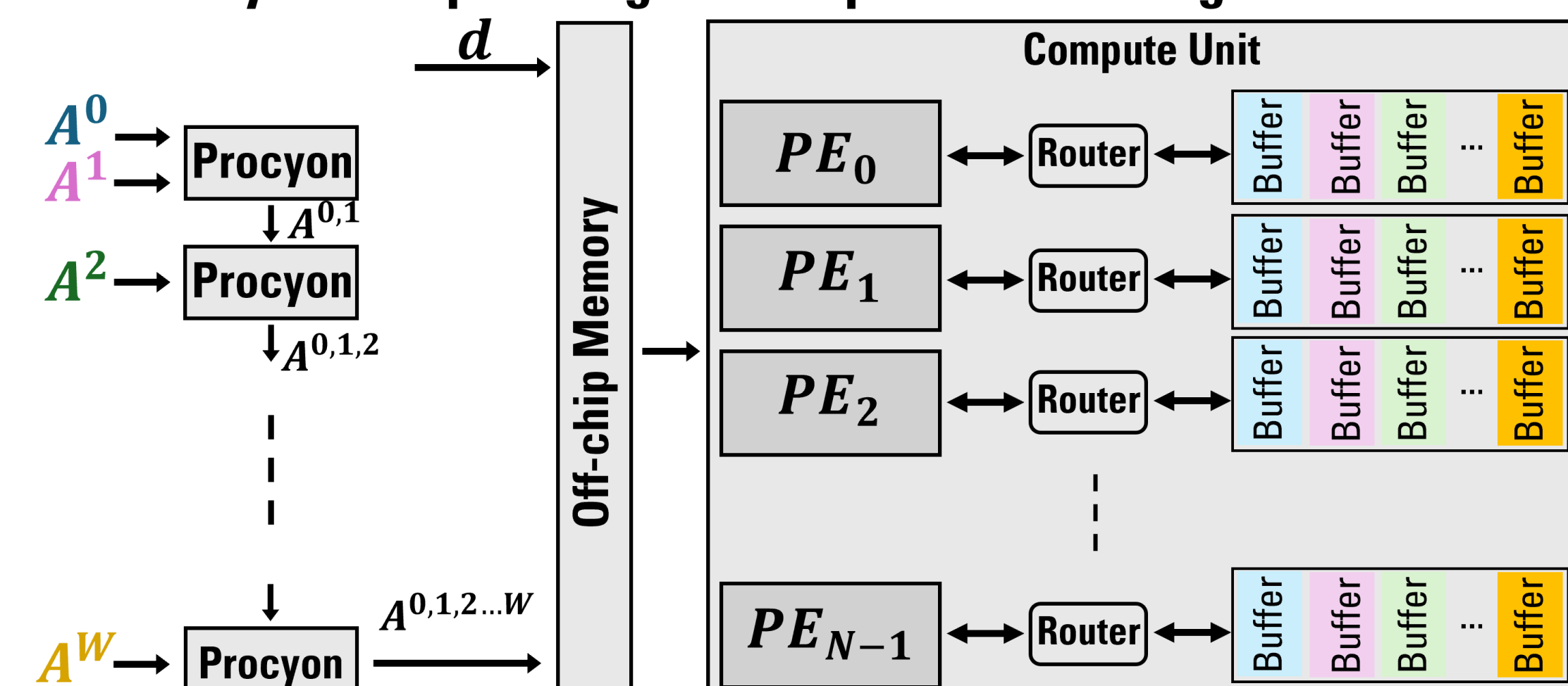


Procyon Architectural Support

a. A general sparse streaming accelerator



b. A Procyon-compatible general sparse streaming accelerator

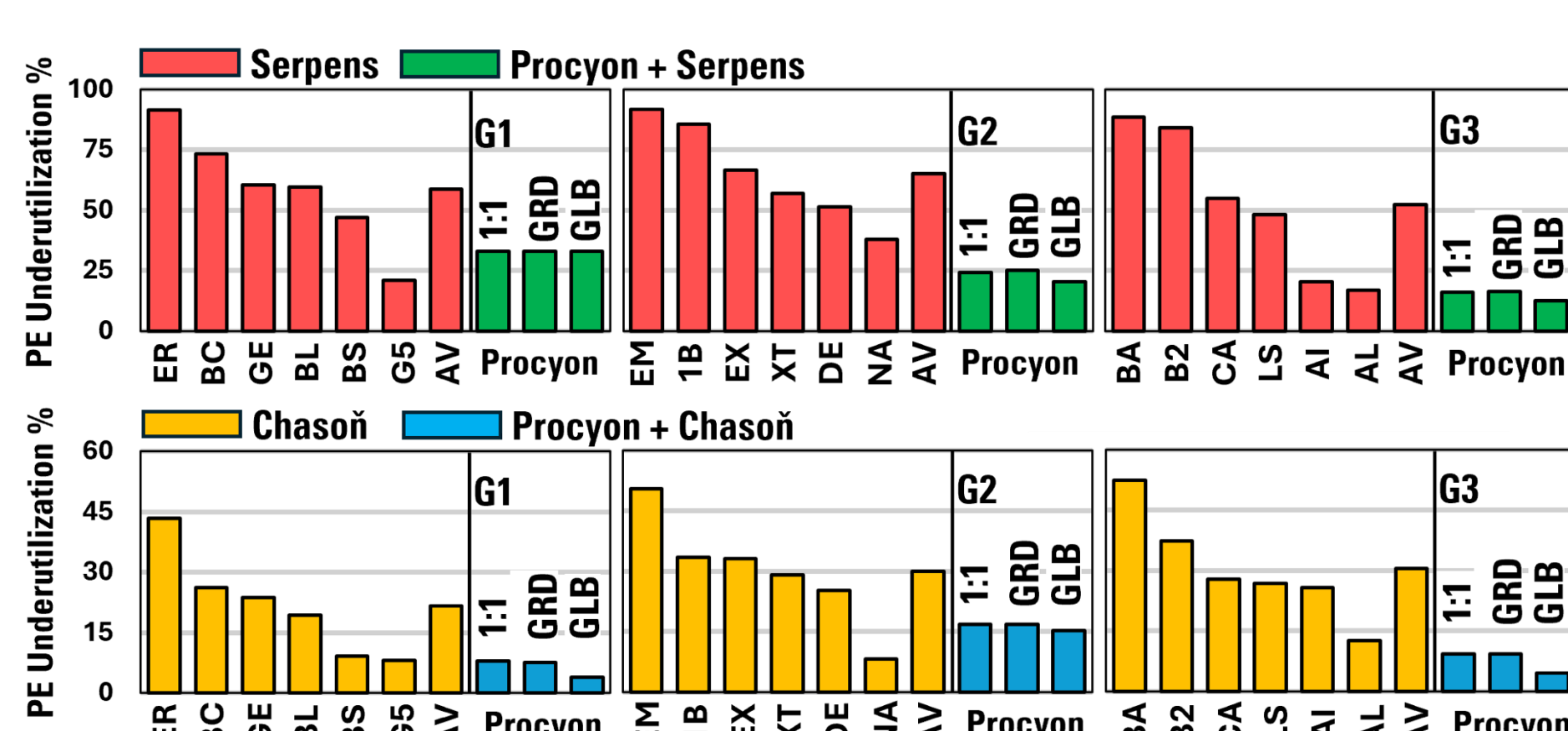


- Multiple workloads ($A^0, A^1, \dots, A^W$) now execute concurrently with the same compute budget of N PEs
- Segregating the output of each workload requires individual scalar units like multiple Merge Units

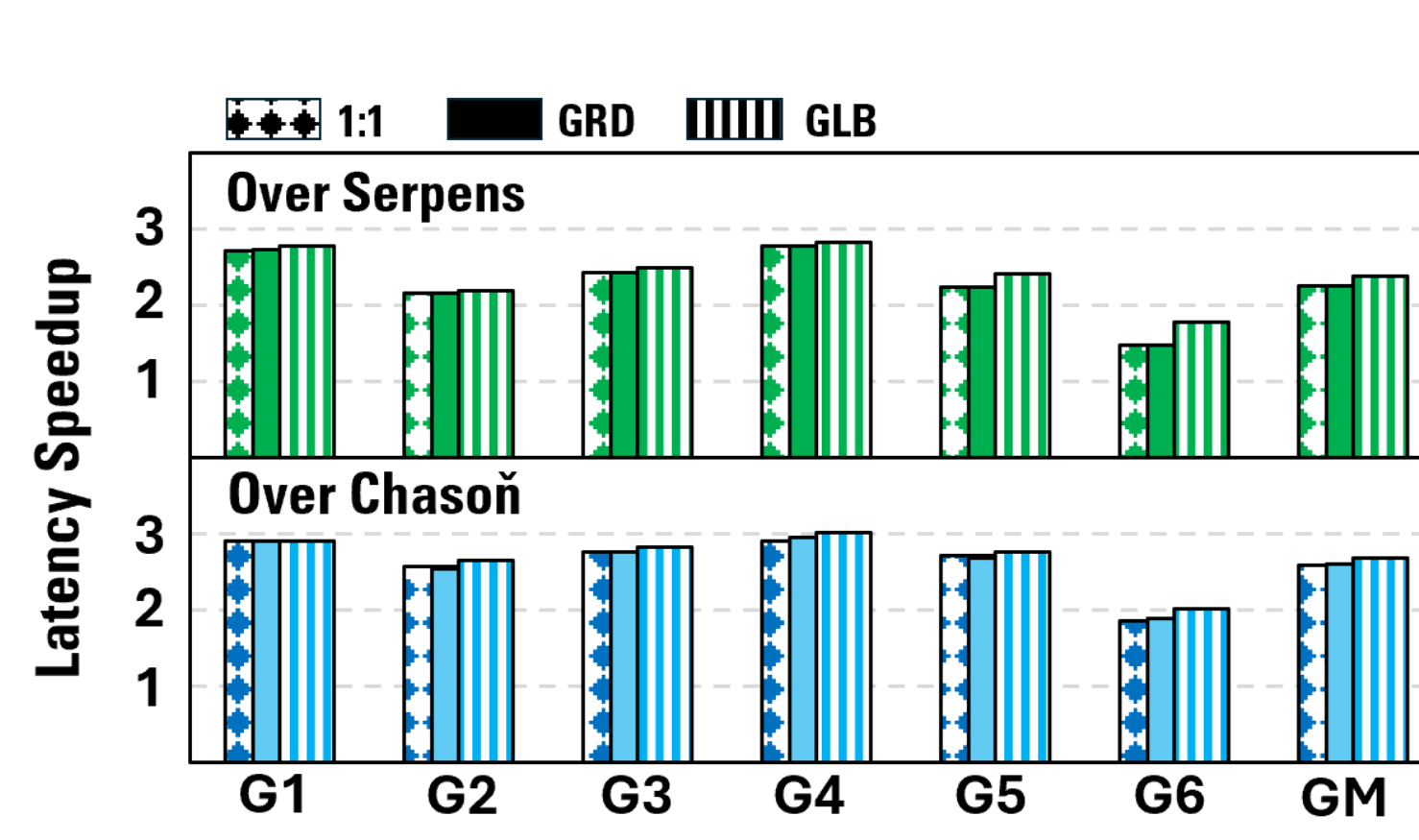
Key Results

- Baselines:** Serpens (DAC 2022) and Chasoñ (MICRO 2025)
- Dataset:** SuiteSparse Workloads
- Sparse kernel:** Sparse-Matrix Vector Multiplication
- Procyon allows a PE datalist to be fused with any other PE datalist, resulting in different PE Pairing schemes. We compare our three PE Pairing schemes namely, 1:1, Greedy, and Global PE Pairing against the baseline.

PE Underutilization. Drops up to $< 10\%$ for Procyon enabled Serpens and Chasoñ



Speedup. Peak $2.9 \times$ and $3 \times$ over Serpens and Chasoñ respectively



Throughput. Procyon+Serpens: peak 49.3 GFLOP/s
Procyon+Chasoñ : peak 61.2 GFLOP/s

